AGRI VISION

Four crops. Four detectors. One shared evaluation split.

A visual demonstration of multi-crop disease detection

Crop disease detection, explained through pictures

Agri Vision detects crop symptoms using a disease label, a confidence score and a bounding box. This booklet presents selected detections and compares four one-stage detectors on shared crop photographs.



Banana



Cauliflower



Groundnut



Radish

How to read the pictures

Green boxes and labels are predictions at confidence 0.25 or higher. Each label identifies a disease or symptom class and the confidence of that detection. The original image on each comparison page is shown as the input reference.

The pictures are selected examples of detector behaviour. Overall performance is measured on the complete validation split, rather than estimated from these illustrations. A confidence score belongs to one detection; it does not describe overall model accuracy.

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A final-year project on federated crop disease detection, combining disease recognition, symptom localisation and comparative detector evaluation.

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Banana | Same photograph, four detectors

Sigatoka



Original photograph

Dataset annotation

Banana sigatoka

Every detector receives the same photograph. The panels display the returned boxes, predicted labels and confidence scores.

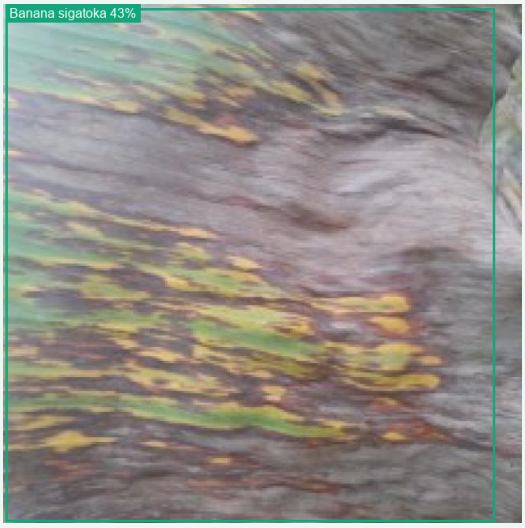
YOLOv8n

Disease labels, bounding boxes and confidence



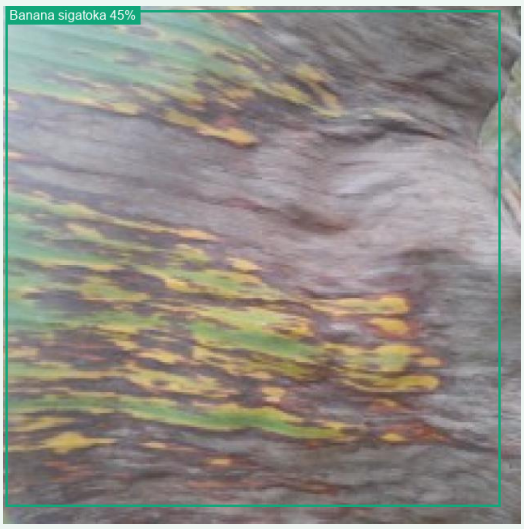
SSDLite

Disease labels, bounding boxes and confidence



SSD-MobileNetV3

Disease labels, bounding boxes and confidence



EfficientDet-D0

Disease labels, bounding boxes and confidence



AGRI VISION

Cauliflower | Same photograph, four detectors

Downy Mildew



Original photograph

Dataset annotation

Cauliflower downy mildew

Every detector receives the same photograph. The panels display the returned boxes, predicted labels and confidence scores.

YOLOv8n

Disease labels, bounding boxes and confidence



SSDLite

No detections above the display threshold



SSD-MobileNetV3

No detections above the display threshold



EfficientDet-D0

Disease labels, bounding boxes and confidence



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Groundnut | Same photograph, four detectors

Rust



Original photograph

Dataset annotation

Groundnut rust

Every detector receives the same photograph. The panels display the returned boxes, predicted labels and confidence scores.

YOLOv8n

Disease labels, bounding boxes and confidence



SSDLite

Disease labels, bounding boxes and confidence



SSD-MobileNetV3

Disease labels, bounding boxes and confidence



EfficientDet-D0

Disease labels, bounding boxes and confidence



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Radish | Same photograph, four detectors

Black Leaf Spot



Original photograph

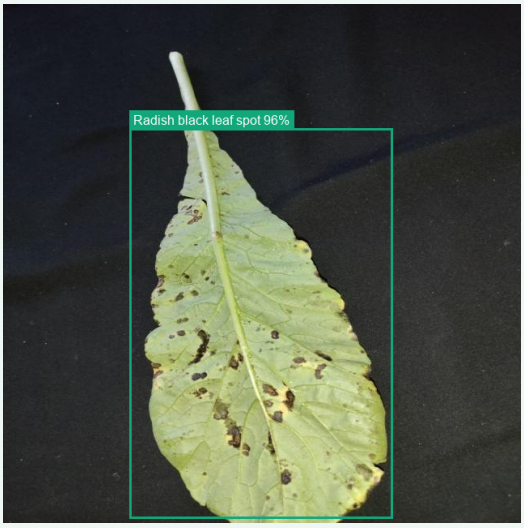
Dataset annotation

Radish black leaf spot

Every detector receives the same photograph. The panels display the returned boxes, predicted labels and confidence scores.

YOLOv8n

Disease labels, bounding boxes and confidence



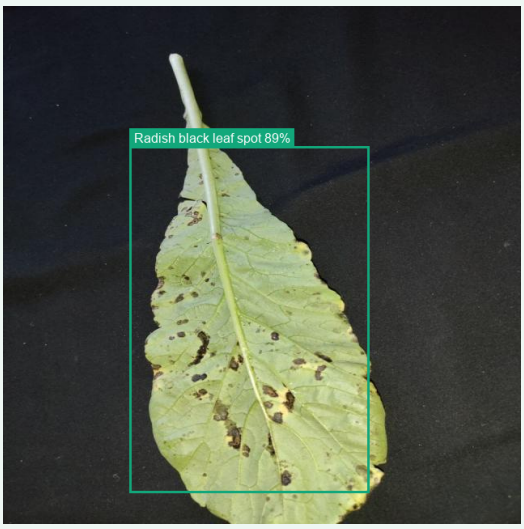
SSDLite

Disease labels, bounding boxes and confidence



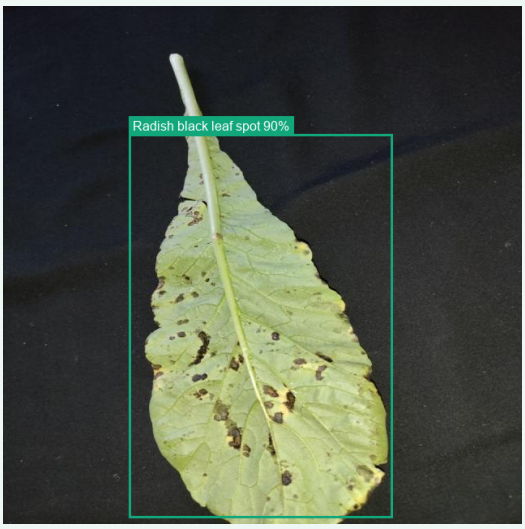
SSD-MobileNetV3

Disease labels, bounding boxes and confidence



EfficientDet-D0

Disease labels, bounding boxes and confidence



Detection performance

Validation results across the shared 24-class dataset

24 classes across banana, cauliflower, groundnut and radish

Shared data: 14,514 training images, 4,157 validation images and 2,091 held-out test images across four crops and 24 classes.

Detector	Precision	Recall	mAP50	mAP50–95
YOLOv8n	60.41%	55.99%	69.47%	45.34%
SSDLite	68.28%	55.84%	65.88%	45.20%
SSD-MobileNetV3	55.52%	55.01%	57.57%	35.20%
EfficientDet-D0	66.05%	78.02%	84.99%	59.56%

Precision measures how many returned detections match the annotated objects. Recall measures how many annotated objects are detected. A useful model should balance both, while localising the affected region accurately.

Precision and recall use the common pooled evaluator at confidence 0.25 and IoU 0.50. AP averages across classes; mAP50–95 tests stricter localization across IoU thresholds 0.50–0.95. Classification accuracy is not substituted for detection AP.

The table reports selected-model validation performance. Held-out test evaluation and consistent inference-speed measurements are planned. The detectors use different input resolutions and architecture-specific training settings.

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Selected crop detections

Strong verified examples across the four crops

Each example below has the correct annotated disease label and a predicted box that overlaps the annotation. Selection considers localisation overlap and confidence.

Banana

YOLOv8n



Cauliflower

YOLOv8n



Groundnut

YOLOv8n



Radish

YOLOv8n



Shared dataset

The study covers banana, cauliflower, groundnut and radish across 24 classes. Fixed training, validation and test splits support consistent evaluation across the detector experiments.

Federated averaging

Three logical clients train locally on separate data partitions. The global model is formed by weighting client updates by their training-image counts. The implementation currently uses custom FedAvg; Flower integration is planned.

Detection outputs

Each detector returns a disease or symptom class, a confidence score and a bounding box. Images in this booklet use actual saved-model inference, with a common display threshold of 0.25.

Next research steps

Evaluate the selected models on the held-out test set, profile inference speed and model size under common hardware conditions, and assess the trade-off between detection quality and deployment cost.

Technologies used

